Chapter 11: **Domination and Hypergraphs**

11.1 The Domination Number of a Graph

A set S of vertices ***dominates*** a graph if every vertex outside the set S is adjacent to at least one member of the set S. The ***domination number*** $\gamma(G)$ of graph G is the minimum size of a dominating set.

For example, for the complete graph K_n we have $\gamma(K_n) = 1$ and for the cycle C_n we have $\gamma(C_n) = \lceil n/3 \rceil$. On a chessboard, the domination problem is related to the minimum number of pieces to attack all squares.

In general we say that each vertex ***dominates*** itself and its neighbors. It follows that each vertex dominates at most $1 + \Delta$ vertices, where Δ denotes the maximum degreee. Thus one obtains the fundamental lower bound:

Theorem. *For a graph G of order n, $\gamma(G) \geq n/(1 + \Delta)$.*

Equality in the bound implies that no two vertices of S are joined by an edge, and every vertex outside S has a unique neighbor in S. This is sometimes called an "efficient/perfect dominating set". For example, two antipodal vertices in the 3-cube Q_3 form such a set. Indeed, in the cube in general, an efficient/perfect dominating set is equivalent to a perfect 1-error-correcting code.

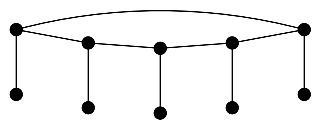
The fundamental upper bound for the domination number is:

Theorem. *If graph G has order n and no isolates, then $\gamma(G) \leq n/2$.*

Proof. A dominating set is ***minimal*** if removing any vertex from it implies that it no longer dominates. Consider a minimal dominating set S. We claim that every vertex of S has a neighbor outside S. Suppose some vertex $v \in S$ has no neighbor outside S. Since the graph has no isolates, v has a neighbor in S. But then when we remove v from S, every vertex is still dominated, a contradiction.

Thus every vertex of S has a neighbor outside S. Or, put another way, the complement of S is also a dominating set of G. This means that G has two disjoint dominating sets, and so one of them has at most half the vertices. ◻

It can be proved that the connected graphs with equality in the above bound are (a) C_4; or (b) the corona of a graph. The ***corona*** of a graph is obtained by adding for each vertex v a new end-vertex v' adjacent only to v. For example, here is the corona of C_5.



The interesting open probem here is Vizing's conjecture for the Cartesian product:

CONJECTURE. *For all graphs G and H it holds that $\gamma(G \square H) \geq \gamma(G)\,\gamma(H)$.*

Another open question is: find the best possible upper bound for the domination number of a cubic (3-regular) graph.

11.2 Nordhaus–Gaddum Results

Early on Nordhaus and Gaddum published a paper about relationships between the chromatic number of a graph and the chromatic number of its complement. In particular they showed that the sum of the chromatic numbers is at most $n+1$ where n is the order. Results of such kind have become known as Nordhaus–Gaddum reusults. Here are the ones for domination, proved by Jaeger and Payan:

THEOREM. *For $n \geq 2$:*
(a) $3 \leq \gamma(G) + \gamma(\bar{G}) \leq n + 1$
(b) $2 \leq \gamma(G) \times \gamma(\bar{G}) \leq n$.

PROOF. Lower bound: except when $n = 1$, it is impossible to have both G and $\bar{G}$ having $\gamma = 1$. Thus one obtains the two bounds.

Upper bound on sum: To get a dominating set of G, take vertex v and its non-neighbors. To get a dominating set of $\bar{G}$, take vertex v and its non-neighbors in $\bar{G}$. So we take v twice and every other vertex at most once. (Attained for complete graph.)

Upper bound on product: Let $S = \{v_1, \ldots, v_\gamma\}$ be a minimum dominating set of G. Each vertex in the graph is dominated by at least one vertex in S.

Thus one can partition $V(G)$ into γ disjoint sets each containing one vertex of S and some of its neighbors. Say the sets in the partition are T_i where T_i contains v_i and is a subset of $N[v_i]$ (the ***closed neighborhood*** of v_i, meaning it and its neighbors). Call a vertex in T_i ***good*** if it dominates T_i. (In other words it can substitute for v_i.) Out of all such partitions, choose the partition such that the number of good vertices is maximized.

Now, consider any of the T_i. We claim that T_i is a dominating set of the complement $\bar{G}$. For suppose it is not. Then there exists vertex w that in $\bar{G}$ is not dominated by T_i. That is, in G, vertex w is adjacent to all of T_i. Say $w \in T_j$. If w is good, then it dominates T_j in G and so $T - \{v_i\}$ dominates G, a contradiction to S being minimum. If w is not good, then this means that moving w to T_i turns it into a good vertex, which contradicts the choice of partition above. In other words, every T_i is a dominating set of $\bar{G}$ and hence has size at least $\gamma(\bar{G})$. It follows that $\gamma \times \gamma(\bar{G}) \leq \sum_{i=1}^{\gamma} |T_i| = n$, and the bound follows.

It is known that there is equality in the product upper bound if and only if G or $\bar{G}$ is: (i) complete; (ii) the union of C_4 and coronas; or (iii) the cartesian product $K_3 \square K_3$. ⌑

11.3 Independent Domination

There are two prominent variations on domination. The first is to require that the dominating set S is an independent set. The independent domination number $i(G)$ is the minimum size of a dominating set of G that is also an independent set. Equivalently, the minimum size of a ***maximal independent set***.

For example: $i(K_n) = 1$; and $i(K_{m,n}) = m$ if $m \leq n$. For the chessboard, we are attacking all squares with mutually nonattacking pieces.

It follows that $i \geq \gamma$. There is no reverse inequality. For example, one can build a graph with $\gamma = 2$ and i arbitrarily large. And there is no nice upper bound; one can build a graph with $i(G) = n - o(n)$.

One nice result is the following, due to Allan and Laskar. Recall that a ***claw*** is an induced copy of $K_{1,3}$.

Theorem. *If graph G is claw-free, then $\gamma(G) = i(G)$.*

Proof. Out of all minimum dominating sets S, take an S such that the number of edges inside S is as small as possible. We claim that that number is zero; that is, S is an independent set.

Suppose that there is an edge inside S, say xy. Then the fact that we need x in S means there is some vertex z outside S that is dominated only by x. (Sometimes called an ***external private neighbor***.)

Let set T be obtained from S by replacing x by z. Note that T has less internal edges than S. The only vertices that T might not dominate are those vertices like z that were originally dominated only by x. But these vertices must form a clique, since otherwise x is the center of a claw with y as one end. Thus z dominates all of these vertices. That is, T is a dominating set of G of the same size as S but with fewer internal edges, a contradiction of the choice of S. That is, S is an independent dominating set. ◻

It is conjectured that for order $n > 10$, a cubic graph has independent domination number at most $3n/8$.

11.4 Total Domination

The other prominent variation of domination is to require that every vertex in S has a neighbor in S. Such a set S is called a ***total dominating set***, and the minimum size is denoted $\gamma_t(G)$.

For example: $\gamma_t(K_n) = 2$; and $\gamma_t(C_n) = n/2$ if n is a multiple of 4. Note that $\gamma \leq \gamma_t$. There is a partial converse in that $\gamma_t \leq 2\gamma$. (Add a mate for every vertex in the minimum dominating set.)

Here the fundamental upper bound is due to Cockayne, Dawes, and Hedetniemi:

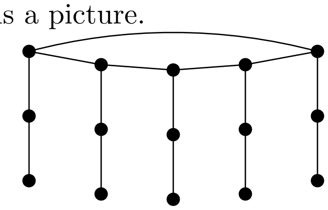
Theorem. *For a graph G of order $n \geq 3$ without isolates, $\gamma_t(G) \leq 2n/3$.*

Proof. We may assume G is a tree, because removing edges can only make one's job harder.

Further, if G has a bridge whose removal leaves two components each with at least 3 vertices, then one can induct. So assume G has no such bridge. In particular, this means that G has diameter at most 4.

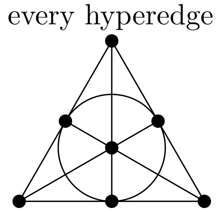
If G has diameter at most 2, then the central vertex dominates, and if diameter 3 then the central pair dominate. In either case $\gamma_t(G) \leq 2$. If G has diameter 4, then by results on trees we know that G has radius 2 and a single central vertex v. It follows that every non leaf-vertex has a leaf neighbor, except possibly v. Thus the set A of non-leaf vertices in G is a total dominating set, and has size at most $(n+1)/2 \leq 2n/3$. ⌑

There is an infinite family of graphs with equality: take a corona and subdivide each leaf edge once. Here is a picture.



11.5 Transversals of Hypergraphs

A ***hypergraph*** has a set V of vertices, and a set E of hyperedges. Each hyperedge is a subset of V. In a graph, every hyperedge has size 2. A famous example of a hypergraph is the Fano plane. This is 3-regular (meaning every vertex is in 3 edges) and 3-uniform (meaning every hyperedge has 3 vertices).



A ***transversal*** of a hypergraph is a set of vertices that intersects every hyperedge. For a graph, a transversal is a vertex cover. Questions about total domination in graphs can be translated into questions about transversals of hypergraphs as follows. The open neighborhood of a vertex in a graph is the set of its neighbors. For a graph G, the ***open-neighborhood hypergraph*** $ONH(G)$ is the hypergraph with the same vertex set as G and such that every open neighborhood in G is a hyperedge in $ONH(G)$. In particular, $ONH(G)$ has n vertices and n hyperedges. If G is r-regular, then $ONH(G)$ is r-uniform.

An interesting result about total domination, found by multiple authors, is that the total domination number of a cubic graph is at most $n/2$ (and there are examples of equality). One way to prove this is to show that the transversal number of a 3-uniform hypergraph with n vertices and n edges is at most $n/2$. This can be done by induction in a cute way, as shown by Thomasse and Yeo.

THEOREM. *A 3-uniform hypergraph has a transversal with size at most $(n+m)/4$ where n is the number of vertices and m the number of hyperedges.*

11.6 Connections

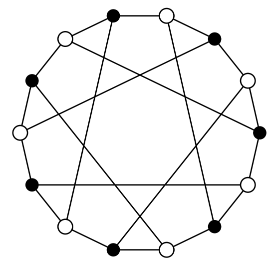
There is a deep connection among (a) set-systems/hypergraphs, (b) bipartite graphs, (c) directed graphs, and (d) matrices.

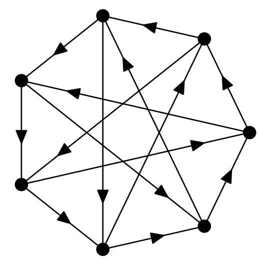
Given a hypergraph, one can construct the ***incidence matrix*** by numbering the hyperedges and vertices, and putting 0 or 1 depending on whether the vertex is in the hyperedge or not. One can then use the incidence matrix as the bipartite adjacency matrix of a bipartite graph: one partite set has a vertex for each row and one partite set has a vertex for each colummn. Note that this process is semi-reversible. If we start with the bipartite graph, then the open-neighborhood hypergraph has two components, one of which is the original hypergraph.

Furthermore, if the resultant bipartite graph has a perfect matching M with bipartition (X, Y), one can form a digraph by the following process. For every edge not in M, orient it from X to Y. Then for every edge in M, contract it to a single vertex. Note that this process is reversible!

Consider for example the Fano plane. One gets the following matrix, and this yields the Heawood graph, and the Paley graph.

$$\begin{bmatrix} 1 & 1 & 0 & 0 & 0 & 1 & 0 \\ 0 & 1 & 1 & 0 & 0 & 0 & 1 \\ 1 & 0 & 1 & 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 1 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 & 1 & 1 & 0 \\ 0 & 0 & 0 & 1 & 0 & 1 & 1 \\ 1 & 0 & 0 & 0 & 1 & 0 & 1 \end{bmatrix}$$





11.7 Disjoint "Dominating" Sets

We saw every graph without isolates has two disjoint dominating sets. There is no minimum-degree condition that guarantees a graph has three disjoint dominating sets, nor two disjoint total dominating sets, nor two disjoint independent dominating sets. If one restricts to regular graphs, then there is such a minimum-degree condition for the first two cases.

One can again connect with hypergraphs. A graph having two disjoint total dominating sets means one can color the vertices red and blue such that every vertex has both a red and a blue neighbor. ***Property B*** for hypergraphs means coloring the vertices with two colors such that every hyperedge contains both colors. Thus the bipartite graph produced from a hypergraph has two disjoint total dominating sets if and only if the original set system has Property B.

It can be shown:

Theorem. *Assume bipartite graph H is regular. Then H has two disjoint total dominating sets if and only if the associated digraph D has an even cycle.*

Thomassen proved that for every $k \geq 3$ every k-regular digraph has an even cycle; and thus for $k \geq 4$ every k-regular bipartite graph has two disjoint total dominating sets.